

Technical Datasheet 4G Gateway

Product Overview: -

The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with Modbus RS485 port and micro Sim compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



Specifications

General –

- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **GPS Location Data:** Provides GPS location data for monitoring of remote equipment.
- **Micro Sim Compatibility:** Supports micro-Sim card for flexible and convenient connectivity option.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Dimensions:** 110 mm L x 110 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.

Configuration Settings: -



- Power up the module by connecting a 12 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 5) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: Wittelb-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage Pages will be accessible.

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	modbus data	Logical Operations
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WiFi Configuration

SSID:

Password:

Device ID:

Data Frequency:

SD Data Upload Time:

Submit

- Enter Device ID (Should be unique for every device).
- Enter data frequency in min to receive data on cloud.
- Enter time to upload SD card data to cloud.

Page 2:- On second page you can set device Baud rate,Parity,Data Bits for RS485 Communication

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	modbus data	Logical Operations
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Rs485 Configuration

Baud Rate:

Data bits:

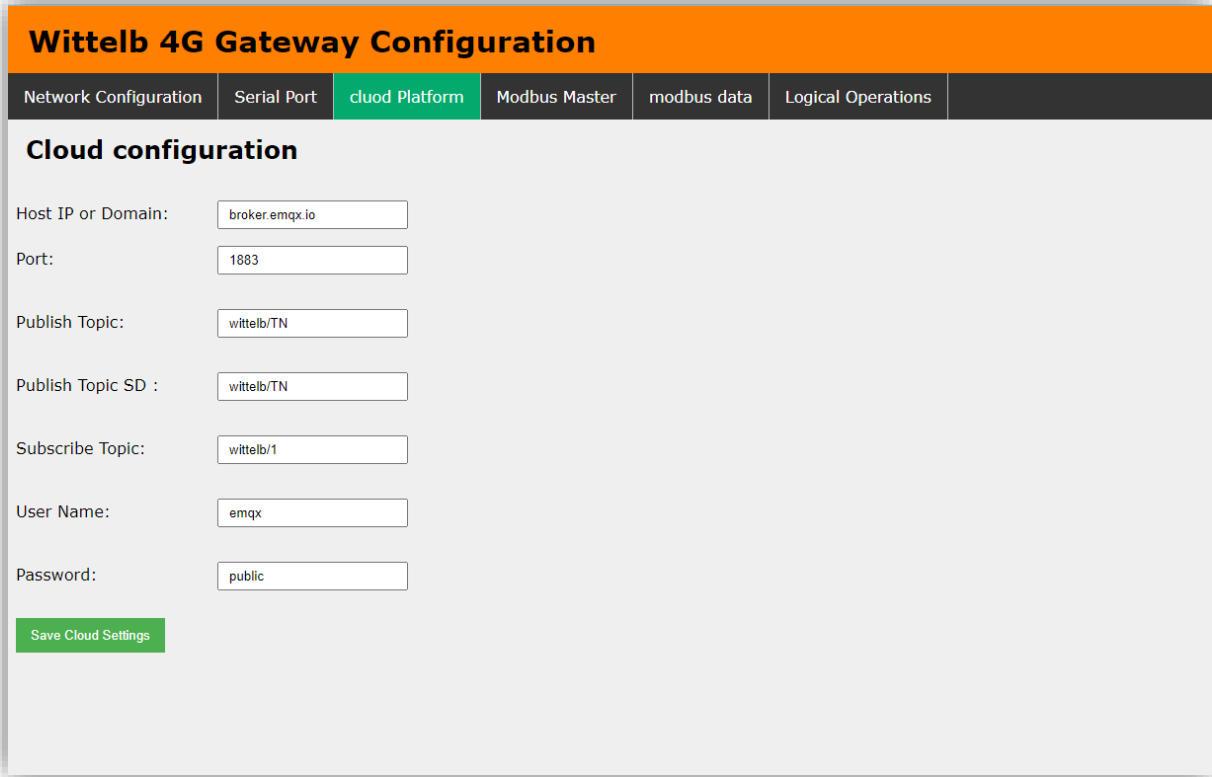
Parity:

Stop Bits:

Save RS485 Configuration

Page 3:-

- Navigate to page 3 on the cloud platform.



Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	modbus data	Logical Operations
Cloud configuration					
Host IP or Domain: broker.emqx.io					
Port: 1883					
Publish Topic: wittelb/TN					
Publish Topic SD : wittelb/TN					
Subscribe Topic: wittelb/1					
User Name: emqx					
Password: public					
Save Cloud Settings					

- Input the Host IP or Domain.
- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.

Page 4:-

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- Indicate the register end address for the slave.
- Enter the function code corresponding to the desired slave.

Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | modbus data | Logical Operations | About

Name	Slave Address	Register start	Quantity	Function Code	Actions
TNH	1	0	2	3	Delete

[Add Row](#) [Submit](#)

Page5:-

- On page 5, you can query each slave to retrieve real-time data.
- You can also rename the register to match your data point name.

Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | Modbus Master | **modbus data** | Logical Operations | About

Select Slave ID:

[Submit](#) [Get Data](#)

Sr.No	Name	Value
1	REG20001	2653
2	REG20002	5500

[Submit](#)

Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | Modbus Master | **modbus data** | Logical Operations

Select Slave ID:

Sr.No	Name	Value
1	REG20001	26.53
2	REG20002	55.00

- On page 6, set the offset value to the actual value.
- Enter the slave ID and specify the desired arithmetic operation.
- Input the register address configured on the previous page.
- Enter the offset value, and the data will be sent to your cloud.

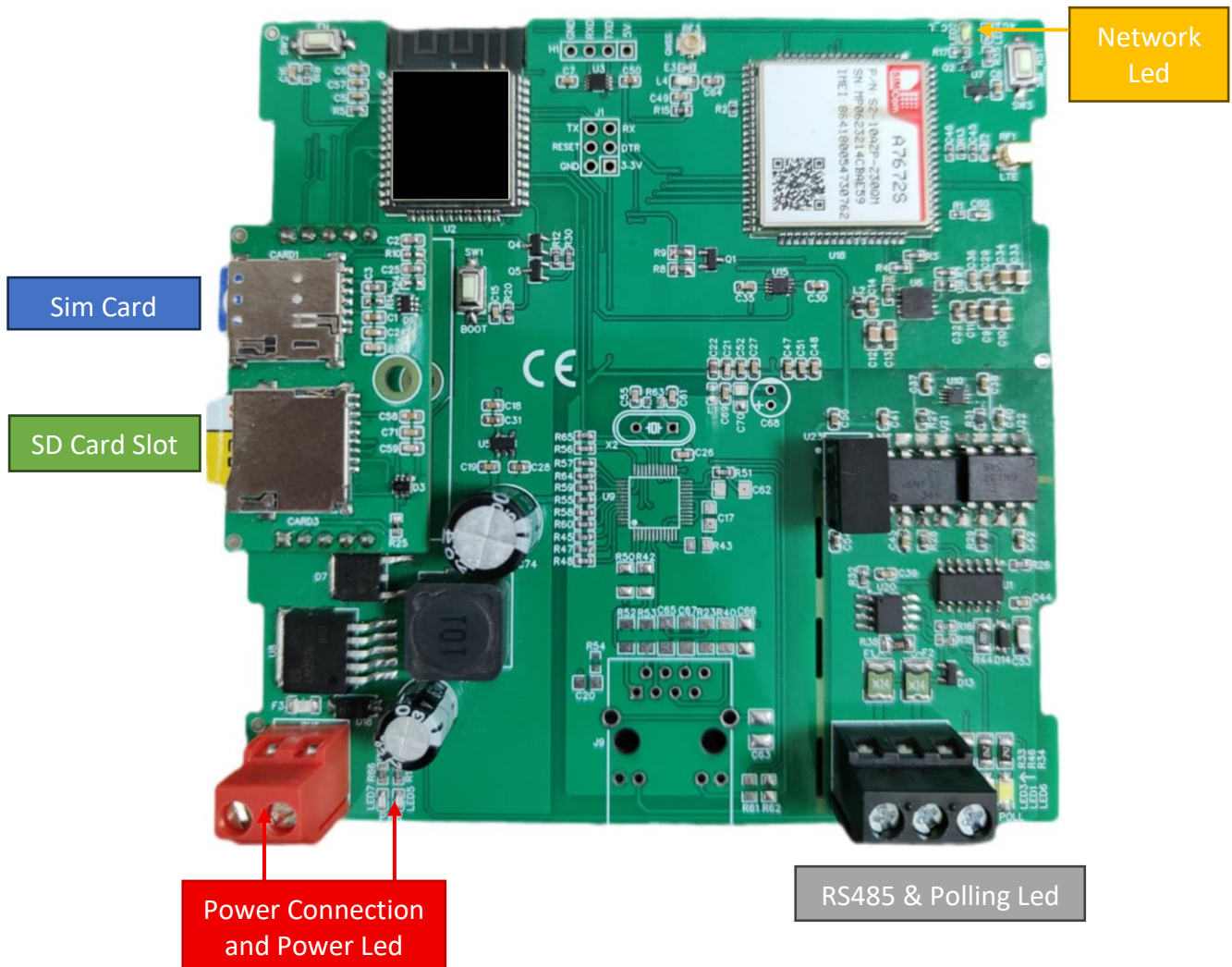
Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | Modbus Master | modbus data | **Logical Operations** | About

Slave Address	Arithmetic Operation	Register Add	Offset Value	Actions
1	Division	REG20001	100	Delete
1	Division	REG20002	100	Delete

- The above image shows data after performing the arithmetic operation with the Offset value applied

4G Gateway connection Diagram: -



Additional Features: -

Communication ports are isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms) 400V isolation.

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